LOCAL COHOMOLOGY OF A NORMALIZED DUALIZING COMPLEX IS THE INJECTIVE HULL IN DEGREE ZERO

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The identification of local cohomology with support in the closed point with the local cohomology of the maximal ideal. Invoked at the start of the counted proof (source line 3651). Rewrites the complex whose cohomology is computed.
- The computation of local cohomology as a homotopy colimit of internal homs out of the quotients by the powers of the ideal. Invoked inside the counted proof (source lines 3656-3657). Exhibits the complex as a homotopy colimit of computable pieces.
- The identification of the internal hom out of a finite length quotient with the corresponding truncation of the injective hull. Invoked inside the counted proof (source lines 3660-3664). Replaces each piece of the homotopy colimit by a truncation of the hull, compatibly with the transition maps.
- The description of the hull as the union of its truncations. Invoked inside the counted proof (source lines 3666-3667). Identifies the colimit with the hull and shows the other cohomology modules vanish.

Source reference. Original lemma lemma-dualizing-finite-length: sources/stacks-dualizing/pinned-original-source.tex, source lines 3204–3204. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing-finite-length, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-local-cohomology-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 1308–1308. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-ext, which

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Source reference. Original lemma lemma-local-cohomology-noetherian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1789–1789. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-union-artinian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1042–1042. Navigation pointer only: the counted text refers to the Stacks item lemma-union-artinian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring. Let $\omega_A^\bullet$ be a normalized dualizing complex. Let $Z = V(\mathfrak{m}) \subset \mathrm{Spec}(A)$. Then $E = R^0\Gamma_Z(\omega_A^\bullet)$ is an injective hull of κ and $R\Gamma_Z(\omega_A^\bullet) = E[0]$.*

Proof. By Lemma lemma-local-cohomology-noetherian we have $R\Gamma_\mathfrak{m} = R\Gamma_Z$. Thus

$$R\Gamma_Z(\omega_A^\bullet) = R\Gamma_\mathfrak{m}(\omega_A^\bullet) = \mathrm{hocolim}\ R\mathrm{Hom}_A(A/\mathfrak{m}^n, \omega_A^\bullet)$$

by Lemma lemma-local-cohomology-ext. Let E' be an injective hull of the residue field. By Lemma lemma-dualizing-finite-length we can find isomorphisms

$$R\mathrm{Hom}_A(A/\mathfrak{m}^n, \omega_A^\bullet) \cong \mathrm{Hom}_A(A/\mathfrak{m}^n, E')[0]$$

compatible with transition maps. Since $E' = \bigcup E'[\mathfrak{m}^n] = \mathrm{colim}\, \mathrm{Hom}_A(A/\mathfrak{m}^n, E')$ by Lemma lemma-union-artinian we conclude that $E \cong E'$ and that all other cohomology groups of the complex $R\Gamma_Z(\omega_A^\bullet)$ are zero. $\square$